SIEM Solution with Malware Analysis and Honeypot: Progress Report and Week 5 Plan

Prepared for the project supervisor. Items marked **[confirm]** are my best reconstruction and should be checked before the meeting.

Weeks 1 to 4: what I did

Week	Focus	What was done
1	Planning	Defined the project: a SIEM with a honeypot, malware analysis and an AI assistant. Studied how QRadar organizes logs, offenses and dashboards. **[confirm]**
2	Honeypot and data	Set up the honeypot and collected sample attack logs and packet captures for testing. **[confirm]**
3	Dashboard	Built the React dashboard with a QRadar-style top bar. Pages: Dashboard, Log Activity, Offenses, Honeypot, Topology, Malware Analysis and AI Assistant. The overview shows a risk gauge, attack charts, a live attack feed and top attackers.
4	AI assistant	Connected the NetSentinel AI agent (Python) to the dashboard. It analyzes packet captures, detects attacks such as port scans and brute force, maps them to MITRE ATT&CK, and answers analyst questions in a chat. Also connected it to the Gemini API.

Week 5 and after Weeks: what I will add

1. **Log database.** Store all logs (time, source IP, event type, severity) so they can be searched and shown on the dashboard instead of sample data.
2. **MITRE ATT&CK dashboard.** A matrix of tactics and techniques, colored by how often each attack was detected. Clicking a technique shows the related alerts.
3. **Login and registration.** Users sign up with name, email and password, then confirm their email before they can log in. Passwords are stored as hashes.
4. **Roles.** SOC analyst, IoT engineer and supervisor. Each role sees only its own pages and data, and the supervisor sees everything.
5. **Login history database.** Save the login time, logout time and time spent for every user. Only the supervisor can see this page.

I will start with the database, because the other four features store their data in it.